

Proves

R4 Cheng

November 7, 2025

Cost function:

$$J(\vec{w}) = \frac{1}{2} \sum_{(\vec{x}, y) \in D} (f_{\vec{w}}(\vec{x}) - y)^2$$

Target:

$$\min_{\vec{w}} J(\vec{w})$$

Gradient Descent

Start from an arbitrary initial weight vector $\vec{w}$.

$$\vec{w}' \leftarrow \vec{w} - \alpha \Delta J(\vec{w}) = \vec{w} - \alpha \frac{\partial}{\partial \vec{w}} J(\vec{w})$$

> α is called learning rate

And for a single training example $(\vec{x}, y)$, we have:

$$\vec{w}' \leftarrow \vec{w} - \alpha(\vec{w} \cdot \vec{x} - y)\vec{x} = \vec{w} + \alpha(y - \vec{w} \cdot \vec{x})\vec{x}$$

Aka. the least mean square (LMS) update rule.

> vs perceptron: $\vec{w}' = \vec{w} + y \cdot \vec{x}$

$$\begin{aligned} \text{Derivation : } \frac{\partial}{\partial \vec{w}} J(\vec{w}) &= \frac{\partial}{\partial \vec{w}} \frac{1}{2} (f(\vec{x}) - y)^2 \\ &= 2 \cdot \frac{1}{2} (f(\vec{x}) - y) \cdot \frac{\partial}{\partial \vec{w}} (f(\vec{x}) - y) \quad // \text{ Chain Rule} \\ &= (\vec{w} \cdot \vec{x} - y) \cdot \frac{\partial}{\partial \vec{w}} (\vec{w} \cdot \vec{x} - y) \\ &= (\vec{w} \cdot \vec{x} - y) \vec{x} \end{aligned}$$